

ECON 562 - Modernization of Bailey (2012)

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1 Synopsis of the Original Paper

In *Reexamining the Impact of Family Planning Programs on US Fertility: Evidence from the War on Poverty and the Early Years of Title X*, Martha Bailey (2012) evaluates the short- and long-term effects of the staggered county-level roll-out of federally funded family planning programs from 1964 to 1973 on U.S. fertility rates.

1.1 Identification Challenge

To evaluate the causal effect of the family planning program on fertility rates, Bailey attempts to recover the average treatment effect on the treated (ATT):

$$ATT = E[Y_{j,t}(1) - Y_{j,t}(0) | D_{j,t} = 1] \quad (1)$$

where $Y_{j,t}(1)$ is the fertility in county j at time t if the county received a program ($D_{j,t} = 1$) and $Y_{j,t}(0)$ is fertility without the program.

The main identification challenge that Bailey faces when evaluating the effect of family planning programs on fertility is that the program placement and timing was not random across counties. So a simple difference in means like in equation (1) would not recover the causal effect due to selection bias. This bias arises because $E[Y_{j,t}(0) | D_{j,t} = 1] \neq E[Y_{j,t}(0) | D_{j,t} = 0]$. That is, because counties had to apply for federal funding and be accepted, meaning that treated and untreated counties would likely have had different fertility outcomes even in the absence of the program. Bailey provides several reasons why this selection bias is a concern.

First, programs were funded through local applications, which implies that local capacity and demand for services drove demand for treatment. Second, treated counties were more likely to be more urban, educated, and wealthy. These characteristics are correlated with fertility behaviour, so even in the absence of the program, treated counties plausibly would have likely experienced different fertility levels and trends. Finally, because the federal initiative explicitly aimed to reduce fertility among disadvantaged women to promote economic opportunities, comparing these early-adopting, wealthier counties to untreated ones conflates the program's causal effect with their pre-existing potential fertility outcomes.

1.2 Methodology

To address this selection bias, Bailey exploits the staggered roll-out of the programs using an event-study specification:

$$Y_{j,t} = \theta_j + \gamma_{s(j),t} + \sum_{y=-7}^{-1} \pi_y D_j \mathbb{1}(t - T_j^* = y) + \sum_{y=1}^{16} \tau_y D_j \mathbb{1}(t - T_j^* = y) + X'_{jt} \beta + \epsilon_{j,t}, \quad (2)$$

where $Y_{j,t}$ is the general fertility rate (GFR; the number of births by county of mother's residence per 1,000 women aged 15 to 44 years), in county j in year t ; θ_j represents county fixed effects; X_{jt} is a vector of county covariates such as socioeconomic and demographic controls interacted with time; D_j is an indicator for ever receiving a family planning grant and T_j^* is the establishment year. The parameter of interest is τ_y , which evaluates the program's effect on fertility y years after it is implemented.

1.2.1 Identifying Assumptions

Bailey's Two-Way Fixed Effect (TWFE) Difference-in-Difference (DiD) design with multiple time periods and variation in treatment timing requires three key assumptions to hold: no anticipation, parallel and common trends, and SUTVA.

1.2.2 No Anticipation

This assumption requires that county residents are not changing their family planning behaviour before a grant is received in a way that impacts their fertility outcomes. Bailey (2012) notes that women in unfunded counties might travel to funded clinics, which would attenuate the estimated results toward zero.

1.2.3 Parallel (Common) Trends

In the absence of the program, treated and untreated counties would have followed the same fertility trends over time. Also, no other shocks should be correlated with treatment timing that affect fertility, for instance counties with high poverty may have received anti-poverty programs (e.g., Medicaid) and family planning programs at the same time, so potential outcomes in fertility could be impacted.

While Bailey shows pre-treatment event-study coefficients are close to zero, the severe covariate imbalance between treated and untreated counties poses a significant threat to this assumption. This will be discussed further in later sections.

1.2.4 Stable Unit Treatment Value Assumption (SUTVA)

A county receiving a family planning program does not affect fertility in another county. A violation would be women going to another county to access funded clinics. No evidence is provided for or against this assumption. If violated, control counties would cause attenuation of the estimated treatment effect of the program toward zero.

2 Relevant Econometric Innovations

Two major econometric innovations developed since 2012 address the limitations in Bailey’s original TWFE estimation.

2.1 Modern DiD Estimators

2.1.1 Motivation

Recent literature demonstrates that TWFE estimators are biased in settings with staggered treatment timing and heterogeneous treatment effects because they implicitly use already-treated units as controls for newly treated units (Goodman-Bacon, 2021; Callaway & Sant’Anna, 2021).

This is highly relevant to Bailey’s study because her treatment effects grow dynamically over time. The family planning program was rolled-out in three distinct policy waves across the United States, so there are three periods, G : the initial programs established between 1965 and 1967, the expansion in 1968 and 1969 as part of the national emphasis program; and programs established from 1970 to 1973 with the start of Title X. Table 3 shows that within three years of the program establishment, the GFR had fallen by over 1 birth per 1,000 women, and by years 6-10, it had fallen by an average of 1.5 births (Bailey, 2012).

Using the Callaway and Sant’Anna (CS) estimator allows us to define potential outcomes based on the time of initial treatment, g . Then, the CS estimator isolates the group-time average treatment effect on the treated

$$ATT(g, t) = E[Y_t(g) - Y_t(0) | G_g = 1]$$

where $Y_t(g)$ is the potential fertility rate at time t if a county was first treated in period g and $Y_t(0)$ is the potential fertility rate in the absence of the program. By strictly comparing newly treated counties only to not-yet-treated or never-treated cohorts, we prevent bias from dynamic treatment effects.

By implementing this innovation, I expect that the estimated treatment effect of the family planning program to be smaller than Bailey (2012)’s results, since the Callaway and Sant’Anna estimator removes already-treated counties from the control group. Since those counties’ fertility rates are declining over time due to the program, their inclusion would lead to an overestimated effect of the program on fertility.

In addition to the no anticipation and SUTVA assumptions outlined in the previous identification section, additional ones are needed for the $ATT(g, t)$ to be valid. First, treatment must be irreversible, which is consistent with how Bailey (2012) describes the program’s implementation. Second, the control group must either use the never-treated or not-yet treated counties, but not already-treated ones. Third, there must be common support (i.e., for each treated county, there is a comparable control county). This assumption will be discussed further in the next section. Finally, the unconditional parallel trends assumption can be relaxed; instead, the conditional parallel trends must hold. This means that conditional on covariates, the average fertility rate for the group first treated in period g and the not-yet-treated or never-treated counties would have followed the same trend in fertility without the family planning program.

2.2 Machine Learning & Covariate Imbalance

2.2.1 Motivation

A key econometric development since Bailey (2012) is the use of flexible, non-parametric estimation methods, such as machine learning, to address covariate imbalance between treated and untreated units. These methods are particularly relevant in settings where treated and control units differ substantially in observable characteristics.

As discussed in the identification section, the DiD strategy requires that parallel trends hold. This requires that, in absence of the family planning program, treated and untreated counties would have experienced similar changes in fertility over time.

As discussed, in Table 1 of Bailey (2012), counties receiving planning programs differ substantially from untreated counties along several dimensions, including income, education, urbanization, and region of residence. These characteristics are correlated with fertility levels as well as changes in fertility over time. As a result, it is plausible that parallel trends do not hold and untreated counties may not be a credible counterfactual for treated counties. For instance, if more urban, educated, and wealthy counties, are more likely to receive treatment are already experiencing decline in their fertility rate over time, then the DiD estimator would over-estimate the effect of the family planning program on fertility.

Bailey (2012) attempts to address this issue by including baseline covariates interacted with time, allowing for differential trends across counties. However, this approach relies on strong functional form assumptions, such as that fertility trends vary linearly with observed characteristics. It is possible there are complex, non-linear relationships between the covariates and fertility that are not accounted by the model. For example, more education, richer, and urban counties may have very different potential outcomes than less educated, poorer, more rural counties.

An alternative approach is to use propensity score weighting. Following the framework of Callaway and Sant’Anna (2021), the propensity score is the probability that a county belongs to a particular treatment cohort given its pre-treatment characteristics. In this setting, inverse probability weighting is used to construct a reweighted control group that resembles the treated group in terms of

observable characteristics. Rather than comparing treated counties to all untreated counties, the method places greater weight on control counties that are more similar to treated counties. As long as untreated and treated counties have a positive chance of being treated or untreated given their covariates (overlap assumption) then we can use these propensity scores to reweigh treated and control counties using inverse probability weights. As a result, the weighted control group provides a more credible estimate of the counterfactual outcome for treated counties, as it helps ensure that treated and control units are comparable in observable characteristics that influence fertility trends. This approach relies on the balancing property of the propensity score (Rosenbaum and Rubin, 1983), which implies that, conditional on the propensity score, the distribution of covariates is the same across treated and control units. In the context of difference-in-differences, this shifts the identifying assumption from unconditional parallel trends to conditional parallel trends, as formalized in Callaway and Sant’Anna, making the assumption more plausible in the presence of substantial covariate imbalance.

Advances in machine learning, such as generalized boosted models (GBM) or random forests, provide a flexible way of estimating these propensity scores. In this setting with staggered treatment timing, one can use GBM to measure these propensity scores for each cohort, which allows for a flexible, non-parametric relationships between covariates and treatment. This flexibility is particularly valuable because mis-specification of the propensity score model can lead to poor covariate balance after weighting. By more accurately estimating the probability of treatment, machine learning methods improve the ability of inverse probability weighting to balance the distribution of covariates between treated and control units. As a result, the reweighted control group more closely resembles the treated group in observable characteristics, making the conditional parallel trends assumption more plausible. In this paper, I use the *R WeightIt* package to estimate propensity scores using GBM, which provides algorithms that better help achieve covariate balance.

2.3 Diagnostics

To assess whether these econometric innovations are relevant to the original paper (Bailey, 2012), several diagnostics are implemented.

Covariate Balance This test helps evaluate whether treated and control counties are comparable in their observable characteristics, X . This balance is needed to provide evidence for the conditional parallel trends assumption, as well as whether there is good enough overlap to use the Callaway and Sant’Anna (2021) DiD estimator.

The choice of covariates used to estimate propensity scores is guided by their relevance for both outcome dynamics and treatment assignment. In particular, covariates should be included if they are likely to influence fertility trends or predict the timing of program adoption. In the context of Bailey (2012), variables such as income, education, urbanization, and demographic composition are especially important, as they are closely related to fertility behavior and are shown in Table 1 to differ substantially between treated and control counties. In addition, including pre-treatment

fertility levels and trends can help account for unobserved factors affecting outcome dynamics.

After estimating propensity scores, covariate balance is assessed using standardized mean differences (SMDs), with successful balance indicated by small differences across all covariates ($SMD < 0.1$). Achieving balance on these variables supports the plausibility of the conditional parallel trends assumption, as it ensures that treated and control counties are comparable along key determinants of fertility trends. However, this is not proof that the assumption holds, rather it is an attempt to make it more credible.

One can estimate propensity scores using never-treated or not-yet-treated counties as the control group. If there is significant imbalance in covariates calculated for the never-treated group, Callaway and Sant’Anna suggest using the not-yet-treated counties instead. This is because the not-yet-treated counties are likely more similar in terms of covariates and propensity scores. However, using this group as the control group requires careful evaluation that the early adopters of the family planning program do not differ in pre-treatment fertility trends from later-treated counties. To do this, I will examine pre-treatment trends across cohorts; if there are significant pre-trends, then I cannot use this group. Failing the test suggests that early adopters differ significantly from later adopters in ways that violate the conditional parallel trends assumptions. Otherwise, if the test passes, we can use not-yet-treated controls as comparison groups.

Overlap It is important to assess whether there is sufficient common support in the distribution of propensity scores between treated and control counties to meet the overlap assumption discussed above (Callaway & Sant’Anna, 2021). In practice, poor overlap occurs when treated counties have propensity scores outside the range of control counties, as well as a large density of propensity scores very close to 0 and 1. Weak overlap would suggest that treated and untreated counties are significantly different in their probability of treatment, and there may be an insufficient amount of control units with similar characteristics to infer what would have happened to treated units in absence of treatment.

Pre-Trends: Pre-treatment event-study coefficients are used to test for differential trends prior to treatment. Significant pre-treatment effects would indicate a violation of the parallel trends assumption. However, the absence of pre-trends is not sufficient to indicate that the parallel trends assumption holds.

Heterogeneous and Dynamic Treatment Effects: To diagnose whether to use the Callaway and Sant’Anna (2021) estimator, we need to show that treatment effects are changing across cohorts and over time. In the context of Bailey (2012), we can see that the family planning program has effects on fertility that is growing. For example, Table 3 (Bailey, 2012) indicates that within three years of program implementation, the general fertility rate declines by approximately 1–1.5 births per 1,000 women of childbearing age, with larger declines of 1.5–2.1 births per 1,000 women observed between six and ten years after implementation. This pattern indicates that treatment effects vary with the length of exposure.

3 Methods and Results

3.1 Methods

The analysis proceeds in three steps. First, baseline covariates are constructed using pre-treatment data, including fertility levels, fertility trends, and socioeconomic characteristics. Second, propensity scores are estimated using GBM to assess balance and overlap under alternative control-group definitions. Third, treatment effects are estimated using the Callaway and Sant’Anna (2021) estimator, with additional robustness checks using alternative estimators. This approach circumvents the negative weighting issues of TWFE by avoiding the use of already-treated counties as effective controls for newly treated counties.

First, to address the severe baseline covariate imbalance between early program adopters and untreated counties, I estimate propensity scores using Generalized Boosted Models (GBM). The model predicts treatment status based on 1960 census socioeconomic characteristics (e.g., urbanization, poverty, nonwhite population) and pre-treatment fertility dynamics (mean fertility levels and linear trends from 1959–1964). I repeat this analysis using never treated counties and not-yet-treated counties to evaluate which comparison group provides the best overlap. The sample is strictly trimmed to the overlapping region of common support to prevent extrapolation outside the bounds of comparable data.

To construct baseline covariates, I use pre-treatment data from 1959–1964. In addition to socioeconomic covariates (income, education, demographic composition, and region indicators), I include two measures of baseline fertility behavior: the mean pre-treatment fertility rate and a county-specific pre-treatment fertility trend. The trend is estimated as the slope from a regression of $Y_{j,t}$ on time using only pre-treatment years. These variables are included to address the possibility that treatment timing is correlated with underlying fertility levels or trends.

To align with the historical rollout of the program, I group counties into three treatment waves as in Bailey (2012). Treated counties in wave g are compared to counties that have not yet been treated at time t .

Finally, I use the Callaway and Sant’Anna (CS) estimator using the GBM-derived weights. The estimator calculates group-time average treatment effects on the treated, $ATT(g, t)$, which are then aggregated into overall, wave-specific, and dynamic event-study parameters. Standard errors are clustered at the county level, and inference is conducted using a multiplier bootstrap procedure.

As a robustness check, I also use two additional DiD estimators and methodologies to address the same issues associated with a staggered treatment timing. I use Sun and Abraham (2021) and a stacked-weighted event study design that explicitly reconstructs the panel into wave-specific, relative-time experiments. Within these stacked local projections, I apply the previously estimated GBM inverse probability weights alongside stack-specific county and year fixed effects.

3.2 Results

3.2.1 Covariate Balance and Overlap Diagnostics

The primary threat to identification in evaluating the staggered roll-out of federal family planning programs is the severe baseline covariate imbalance between control and treated counties. To assess the validity of the control group, I first evaluated the use of a "never-treated" counterfactual pool. As demonstrated in Table 1 and visualized in Figure 2 (Panel A), comparing the earliest program adopters (Wave 1: 1965–1967) to never-treated counties yields imbalances outside of the acceptable range. Even after the application of Generalized Boosted Model (GBM) propensity score weights, the maximum absolute Standardized Mean Difference (SMD) remains high at 0.596, with 11 baseline covariates failing to meet the conventional $SMD \leq 0.10$ threshold.

To construct a more credible counterfactual, I restrict the donor pool strictly to not-yet-treated counties. Table 1 shows that, when combined with GBM weighting and trimmed to the region of common support, comparability between treated and not-yet-treated counties improves a lot. For the program's primary expansion phase (Wave 2: 1968–1969), the mean absolute SMD falls to 0.127, with the majority of covariates achieving balance. This improvement is visually confirmed in Figure 2 (Panel B), which shows the post-weighting SMDs clustering much closer to the zero-line for the expansion waves. While Wave 1 remains challenging to fully balance due to the extreme population scale of the initial urban grantees (Max SMD = 0.852), the use of not-yet-treated controls helps provide more credibility for the conditional parallel trends assumption.

Furthermore, I analyze pre-treatment fertility trends, which provides evidence that the treatment cohorts were on similar trajectories prior to the program. A regression of the county-specific 1959–1964 fertility trend on treatment wave indicators yields statistically insignificant coefficients for both the Wave 2 indicator ($p=0.729$) and the Wave 3 indicator ($p=0.651$), failing to reject the null hypothesis of equal pre-trends relative to Wave 1. However this is not definitive proof that the parallel trends hold.

3.2.2 Pre-Treatment Trends

To formally test the conditional parallel trends assumption, I extract the pre-treatment event-study coefficients from the dynamic aggregation of the Callaway and Sant'Anna (2021) estimator. Table 3 reports these placebo estimates for the six years prior to program adoption.

The results strongly support the validity of the research design. For all pre-treatment periods $t \in \{-6, \dots, -2\}$, the point estimates are small and fail to reach statistical significance at the 5% level. For example, the estimated effect three years prior to treatment ($t=-3$) is 0.190 births per 1,000 women ($SE=0.418$). This absence of significant pre-treatment effects, provides evidence against anticipation behavior and validates the use of the reweighted, not-yet-treated counties as a credible counterfactual.

3.2.3 Treatment Effects

The primary causal estimates from the Callaway and Sant’Anna doubly-robust estimator demonstrate that federally funded family planning programs induced a meaningful reduction in the General Fertility Rate (GFR). The overall Average Treatment Effect on the Treated (ATT) across all cohorts and post-treatment periods is 1.111 births per 1,000 women aged 15–44 (SE=0.621), representing a marginally significant decline ($p < 0.10$).

Decomposing the effects by treatment cohort (reported in Table 2) reveals substantial heterogeneity in the program’s impact. The strongest and most precisely estimated reduction occurred during the Wave 2 expansion. Specifically, the 1968 treatment cohort experienced a highly significant ATT of 4.482 (SE=1.512, 95% CI: [7.446, 1.518]). Similarly, within Wave 3, the 1971 treatment cohort exhibited a significant decline of 2.736 births per 1,000 women (SE=1.221). While the point estimates for the initial Wave 1 cohorts (e.g., 1965: 3.989) are also large and negative, they are estimated with wide standard errors (SE=2.561) due to the aforementioned residual covariate imbalance, and thus do not achieve statistical significance at conventional levels.

The dynamic event-study aggregation, plotted in Figure 3, illustrates that the treatment effects are not instantaneous. Following the reference year ($t=1$), the point estimates initially hover near zero before beginning a sustained, accelerating decline starting around event year $t=2$. This dynamic trajectory is consistent with the institutional lag required to establish functional clinics and the biological lag inherent in preventing pregnancies.

3.2.4 Alternative Estimators

To ensure that the observed fertility declines are not an artifact of using the Callaway-Sant’Anna estimator, I estimate the treatment effects designed for staggered rollout, including a stacked estimation design using the GBM weights and the Sun and Abraham (2021) estimator. As plotted in Figure 5, the negative treatment effects are consistent across all three identification strategies. The interaction-weighted estimator (Sun and Abraham, 2021) and the Stacked-Weighted event study both report statistically insignificant, flat pre-trends, further validating the parallel trends assumption. More importantly, all three models detect an increasing decline in the GFR in the post-treatment period. However, the estimates are noisy due to the smaller sample sizes used to estimate the effects.

4 Discussion

This modernization provides rigorous confirmation of Bailey’s (2012) core conclusion: the rollout of federal family planning programs during the War on Poverty caused a measurable and robust decline in U.S. fertility rates. However, subjecting the original observational data to frontier econometric techniques reveals critical nuances regarding the magnitude of this effect and the extreme sensitivity of traditional estimates to control group selection. Bailey’s original Two-Way Fixed Effects (TWFE)

specification estimated that the programs reduced the General Fertility Rate by approximately 1.5 to 2.1 births per 1,000 women six to ten years after establishment. In contrast, our doubly-robust overall ATT is slightly attenuated at 1.111 births per 1,000 women. By restricting the control group to not-yet-treated counties, the modernized Callaway and Sant’Anna (2021) estimator successfully removes this dynamic treatment effect bias.

Furthermore, this analysis isolates and highlights the severe “first-mover” selection bias that inherently plagues evaluations of historical federal grants. The persistent covariate imbalance in Wave 1 demonstrates that the earliest adopting counties were extreme demographic outliers that lack perfect counterfactuals even when applying advanced Generalized Boosted Models (GBM) to a restricted donor pool. Because perfect conditional parallel trends cannot be guaranteed for these initial 1965–1967 grantees, their estimated treatment effects remain highly noisy. Consequently, the modernization succeeds most clearly in evaluating the program’s Wave 2 expansion (1968–1969). During this period, the federal initiative reached a broader, more representative set of counties, allowing the GBM model to achieve excellent covariate balance. It is precisely within this cleanly matched quasi-experiment that the data reveals a highly significant and dramatic reduction of 4.48 births per 1,000 women.

Ultimately, the consistency of the dynamic treatment effects across the Callaway-Sant’Anna, Sun-Abraham, and Stacked-Weighted estimators shows that the reduction in fertility is stable estimate. While residual imbalances in the earliest cohorts necessitate a cautious interpretation of the initial roll-out, this modernized approach successfully validates the historical efficacy of the policy. The federal family planning initiatives induced a decline in fertility that grew progressively stronger as the clinics became established in their communities.

5 References

- Bailey, M. J. (2012). Reexamining the Impact of Family Planning Programs on U.S. Fertility: Evidence from the War on Poverty and the Early Years of Title X. *American Economic Journal: Applied Economics*, 4(2), 62-97. <https://doi.org/10.1257/app.4.2.62>
- Callaway, B., & Sant’Anna, P. H. C. (2021). Difference-in-Differences with multiple time periods. *Journal of Econometrics*, 225(2), 200-230. <https://doi.org/10.1016/j.jeconom.2020.12.001>
- Crump, R. K., Hotz, V. J., Imbens, G. W., & Mitnik, O. A. (2009). Dealing with limited overlap in estimation of average treatment effects. *Biometrika*, 96(1), 189-199. <https://doi.org/10.1093/biomet/asn055>
- Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2), 175-199. <https://doi.org/10.1016/j.jeconom.2020.12.001>

6 Appendix

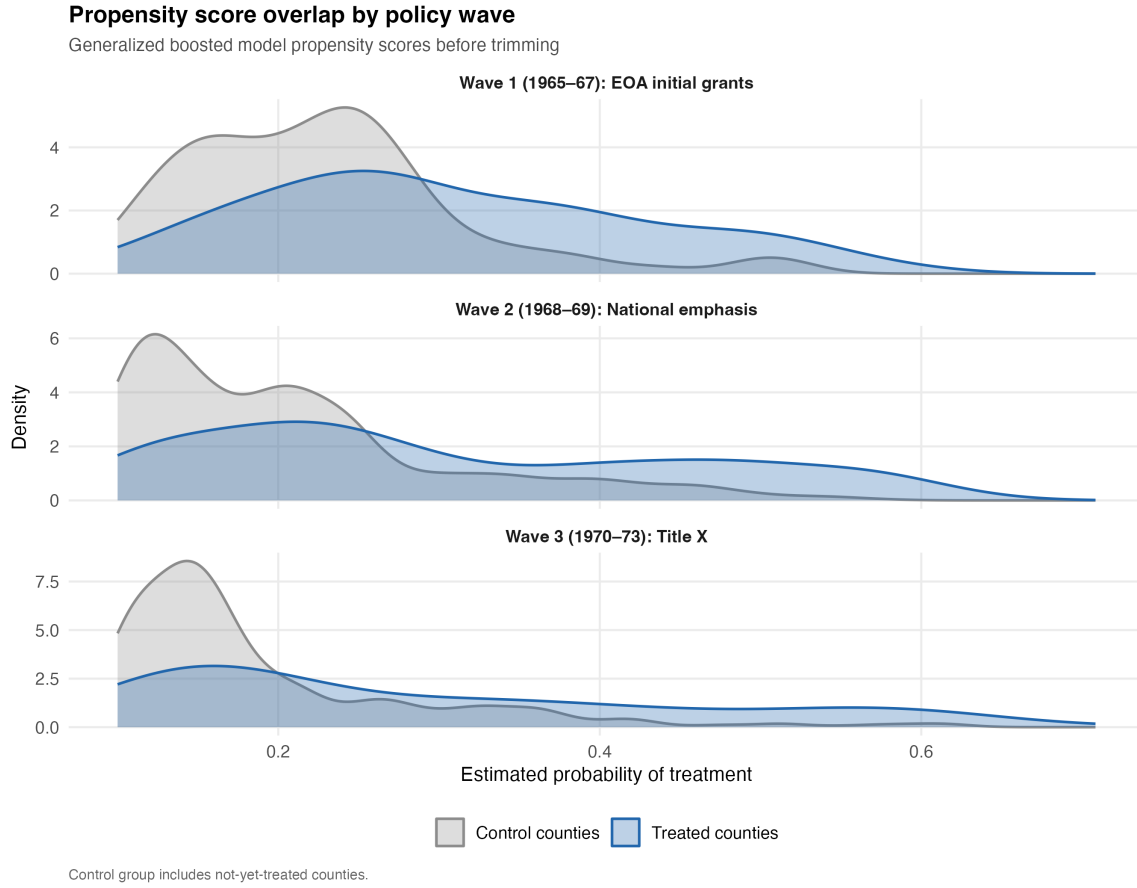


Figure 1: **Propensity Score Overlap by Policy Wave.** This figure displays the distribution of estimated propensity scores (the probability of treatment conditional on 1960 baseline characteristics and pre-treatment fertility trends) for treated counties and not-yet-treated control counties across the three policy waves. The densities are calculated prior to common-support trimming. The presence of overlapping regions justifies the use of inverse probability weighting, although the skewness in Wave 1 highlights the fundamental differences between early adopters and the available control pool.

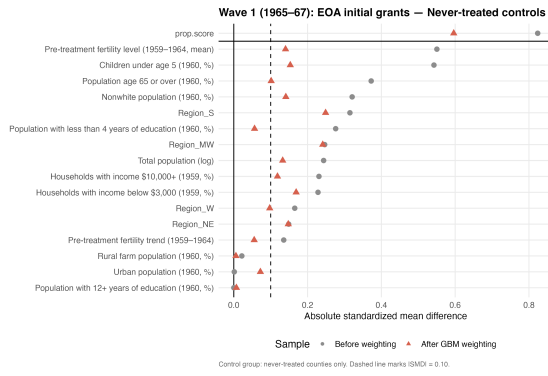
Table 1: Covariate Balance Across Alternative Control Group Designs

Policy	Wave	Max SMD (Unweighted) ^a	Never-Treated Controls		Not-Yet-Treated Controls			
			Max SMD	$N > 0.10^b$	Max SMD	Mean SMD	$N > 0.10^b$	$N > 0.25^b$
Wave (1965–67): EOA grants	1	0.824	0.596	11	0.852	0.119	6	1
Wave (1968–69): National emphasis	2	0.600	0.592	4	0.622	0.127	5	2
Wave (1970–73): Title X	3	0.901	0.657	1	0.983	0.158	8	2

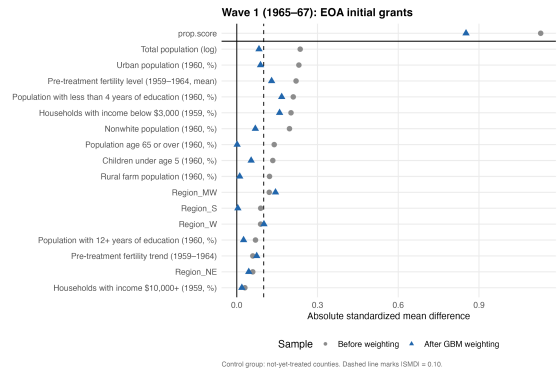
Note: Standardized Mean Differences (SMDs) are computed after Generalized Boosted Model (GBM) propensity score weighting. A lower SMD indicates better covariate balance between the treated and control counties. The sample is trimmed to the overlapping region of common support to improve comparability.

^a Maximum absolute SMD across all covariates before any propensity score weighting is applied.

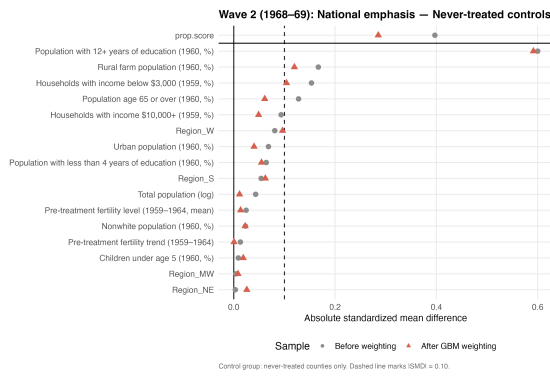
^b Number of covariates with an absolute SMD exceeding the specified threshold (0.10 is the conventional target for strong balance).



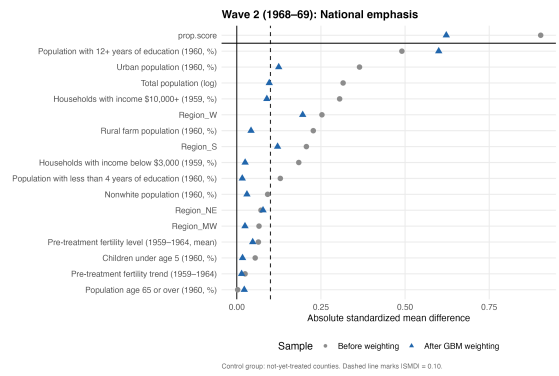
(a) Wave 1: Never-Treated



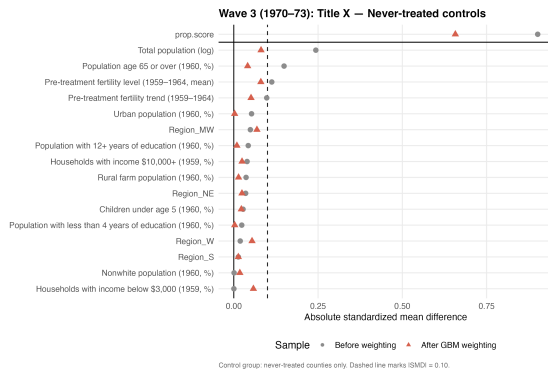
(b) Wave 1: Not-Yet-Treated



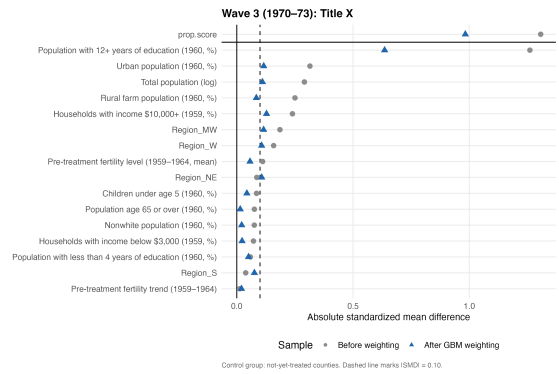
(c) Wave 2: Never-Treated



(d) Wave 2: Not-Yet-Treated



(e) Wave 3: Never-Treated



(f) Wave 3: Not-Yet-Treated

Figure 2: **Covariate Balance Before and After GBM Propensity Score Weighting.** This figure compares the absolute standardized mean differences (SMDs) of 1960 baseline covariates and pre-treatment fertility trends across the three policy waves. The **left column** displays the balance when using a static pool of *never-treated* counties as the control group. The **right column** displays the balance achieved by restricting the control group to *not-yet-treated* counties. Grey circles represent the unweighted, naive differences. Blue triangles represent the differences after applying Generalized Boosted Model (GBM) propensity score weights. The dashed vertical line marks the conventional $|SMD| = 0.10$ threshold for acceptable balance. By comparing the left and right columns row-by-row, it is evident that utilizing a not-yet-treated control group significantly improves post-weighting covariate balance for the expansion waves (1968 and 1970).

Table 2: Average Treatment Effects of Family Planning on Fertility

<i>Dependent Variable: General Fertility Rate</i>	
Overall Average Treatment Effect (ATT)	−1.111* (0.579)
<i>Independent Variable: Treatment by Policy Wave</i>	
Wave 1 (1965–1967 Cohort)	−3.989 (2.561)
Treated in 1966	−2.168 (2.209)
Treated in 1967	1.349 (1.576)
Wave 2 (1968–1969 Cohort)	−4.482*** (1.512)
Treated in 1969	−0.639 (0.765)
Wave 3 (1970–1973 Cohort)	−1.218 (1.523)
Treated in 1971	−2.736** (1.221)
Treated in 1972	0.232 (1.075)
Treated in 1973	−3.488 (3.028)
Observations (County-Years)	91,110
Control Group	Not-Yet-Treated
Estimator	Callaway-Sant’Anna (2021)

Notes: Estimates are generated using the Callaway and Sant’Anna (2021) doubly-robust estimator with a not-yet-treated control group. Note that while GBM weights were used for sample diagnostics and robustness checks, this primary CS specification calculates its own internal propensity scores and outcome regressions to achieve the doubly-robust property.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 3: Pre-Treatment Event-Study Estimates (Placebo Test)

<i>Dependent Variable: General Fertility Rate</i>	
<i>Independent Variable: Years Before Treatment</i>	
$t = -6$	0.106 (0.491)
$t = -5$	0.109 (0.451)
$t = -4$	-0.212 (0.410)
$t = -3$	-0.190 (0.418)
$t = -2$	-0.158 (0.388)
$t = -1$	(Omitted Reference Period)
Control Group	Not-Yet-Treated
Estimator	Callaway-Sant'Anna (2021)

Notes: Standard errors clustered at the county level are reported in parentheses. The dependent variable is the General Fertility Rate. The independent variables are event-time indicators representing the years prior to the initial federal family planning grant. The period $t = -1$ serves as the normalized reference period.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

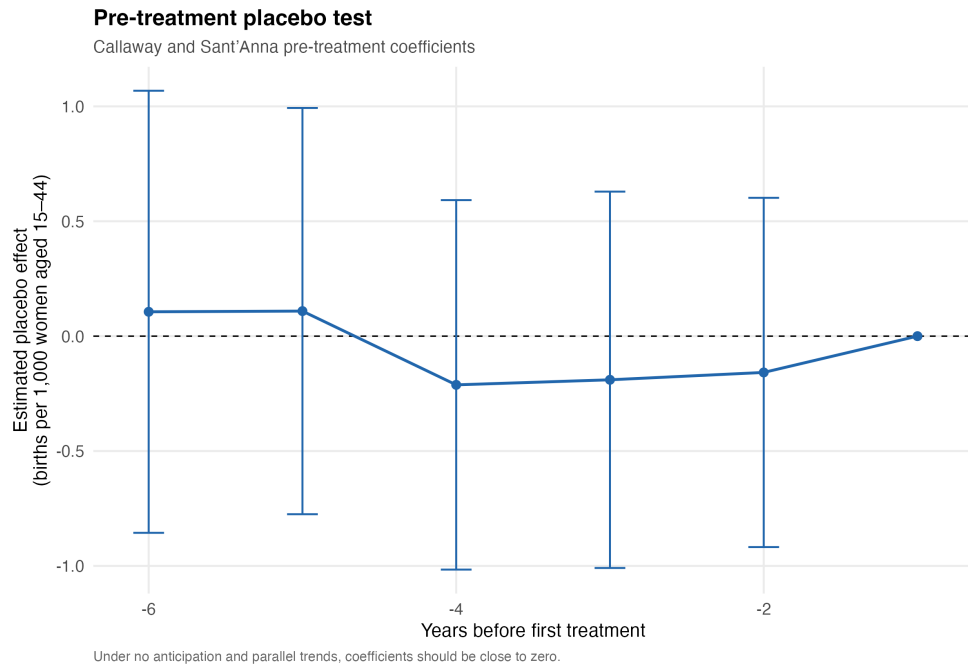


Figure 3: Pre-Treatment Placebo Test. This figure plots the Callaway and Sant'Anna (2021) pre-treatment event-study coefficients. The x-axis represents the number of years prior to the receipt of the first federal family planning grant. If the conditional parallel trends assumption holds, these placebo estimates should be close to zero. As shown, all pre-period coefficients are statistically insignificant and tightly centered around zero, providing strong evidence against anticipation effects and supporting the validity of the identification strategy.

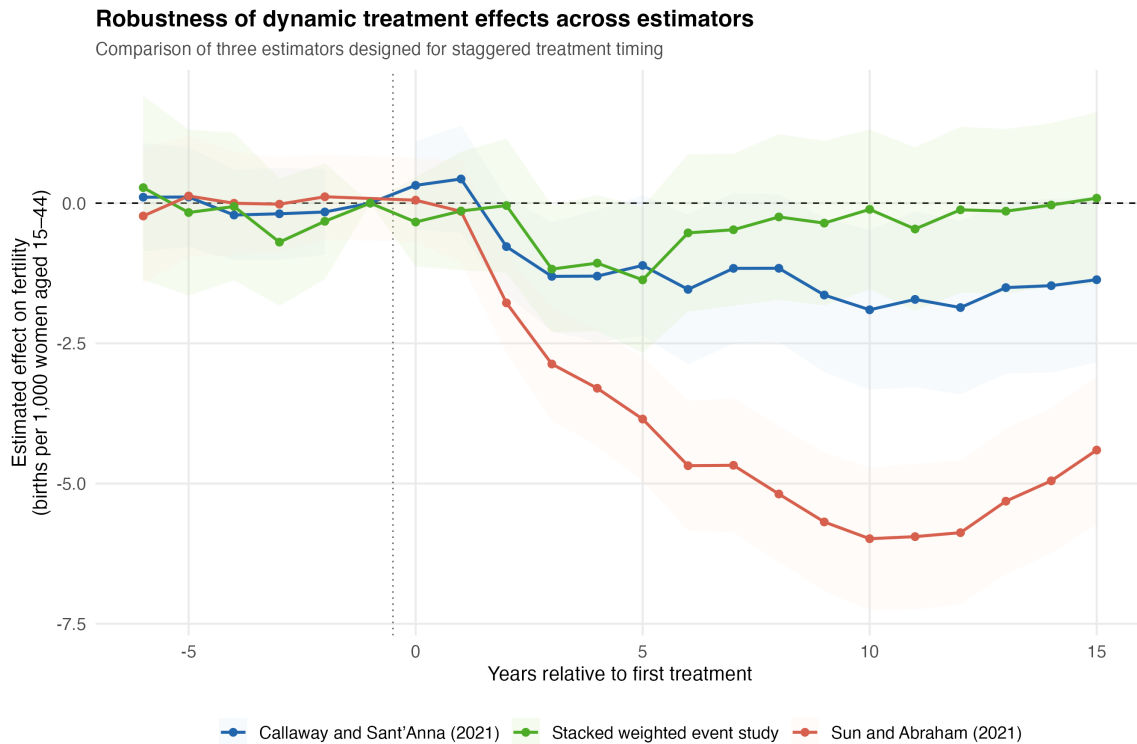


Figure 4: **Dynamic Treatment Effects and Estimator Robustness.** This figure presents the estimated dynamic effects of federal family planning programs on the general fertility rate (births per 1,000 women aged 15–44) across three modern staggered difference-in-differences estimators: Callaway and Sant'Anna (2021), Sun and Abraham (2021), and a stacked weighted event study using GBM weights. The consistent, deepening negative trajectory across all three specifications confirms that the programs led to a robust and significant reduction in fertility over time.